Andes Virus Outbreak

MV Hondius Cruise Ship · 2026

29 May 2026

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Outbreak summary · 29 May 2026
13 cases in analysis (11 confirmed · 2 probable) · 3 deaths · 03 Apr – 25 May 2026

Background

Andes virus (ANDV) is the only hantavirus documented to transmit person-to-person, in addition to the usual rodent-to-human route (Alonso et al, 2020). The 2026 outbreak is linked to the **MV Hondius** expedition cruise ship. Cases first appeared in early April 2026. By the time whole-genome sequencing confirmed ANDV as the causative agent, passengers and crew had dispersed across Europe, Asia, and the Americas.

Methods

Data. All cases classified as *confirmed* or *probable* (13 total) were included in the analysis. For each case we used a single reference date, defined as the earliest available of symptom onset, laboratory confirmation, or hospitalisation (see Note). L-segment sequences were retrieved from Pathoplexus for the 4 of 13 cases with deposited accessions. Cases with a recorded `ship_board_date` contribute a timed-contact entry to the model, representing the ship as a shared exposure setting between boarding and the earliest of disembarkation or death. Cases 23 and 24 lack both `ship_board_date` and ship_disembark_date in the linelist; we manually added them to the contact data with an exposure window spanning the full cohort range.

Model. Transmission trees were inferred using outbreaker2, a Bayesian inference framework that infers transmission chains from epidemiological, contact, and genetic data (Jombart et al, 2014; Campbell et al, 2018). We assumed the serial interval as a discretised Gamma distribution with mean 18 days and CV = 1/3 (UKHSA rapid report, 2026). Up to two unobserved intermediate generations were allowed between observed cases (max_kappa = 3). The reporting rate π was assumed to follow a Beta(10, 1) prior (initial value 0.95), reflecting the expectation that case ascertainment on a closed cruise cohort is high. The contact reporting coverage (ε) and place persistence (τ) were both fixed at 1, reflecting the assumption that recorded ship contacts are complete and that the ship was the sole shared transmission setting during the voyage. Because the L-segment alignment contains near-zero genetic variation (see SNP matrix in Results), the per-site mutation rate μ is non-identifiable from these data and was fixed at 1.7×10^{-5} substitutions/site/generation, derived from the published ANDV substitution rate of $\sim 3.5 \times 10^{-4}$ subs/site/year (Ramsden et al, 2008) and an 18-day serial interval.

Inference. Four MCMC chains were run independently for 100,000 iterations each, thinned every 50 steps; the first 10,000 iterations per chain were discarded as burn-in. Convergence was assessed by Gelman–Rubin $\hat{R}$ and effective sample size across chains (see Diagnostics). Consensus ancestries were taken as each case's modal inferred infector across the pooled posterior; ancestry uncertainty is summarised by the Shannon entropy (o2ools package).

Results

Timeline of Exposure Windows

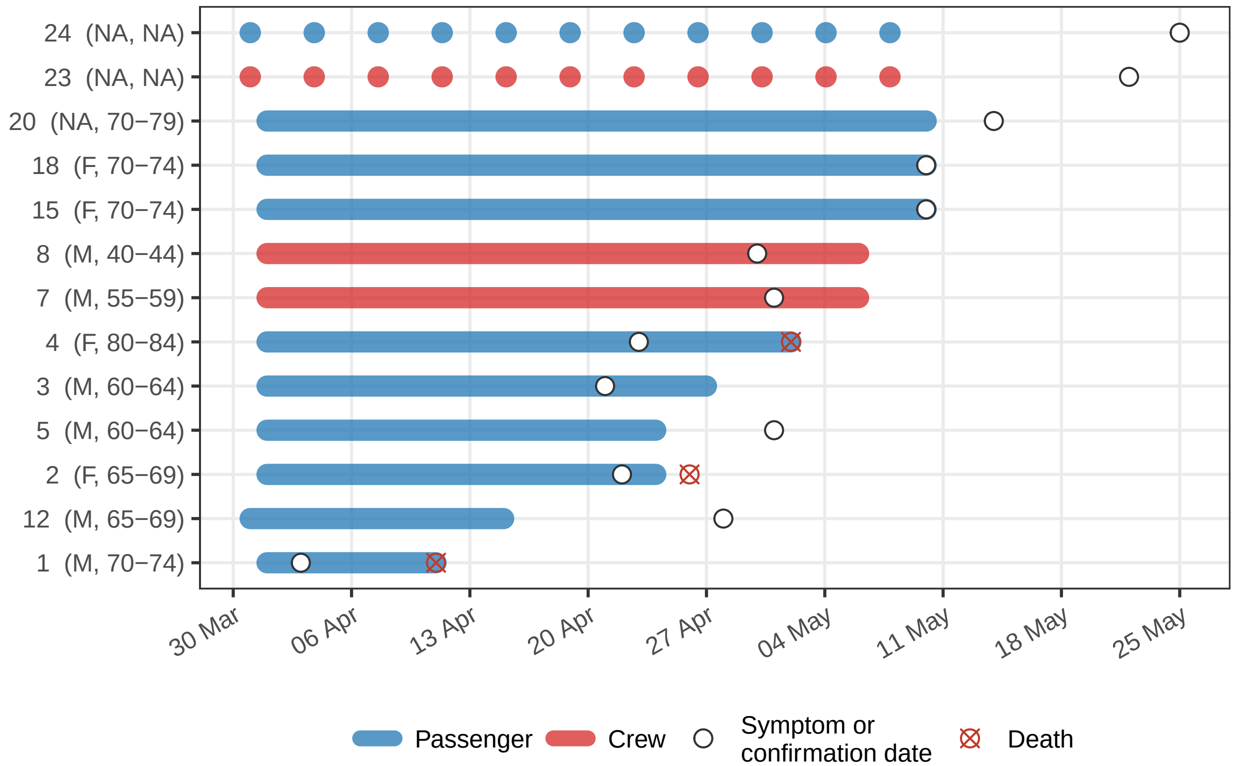

Figure 1: Exposure windows per case defined by time spent aboard the MV Hondius. Segments represent the time periods during which each individual was present on the ship, with colours representing role. Circles indicate symptom onset and X represents death. Dotted segments (cases 23 and 24) denote windows imputed to the full cohort exposure range.

Genomic Analysis

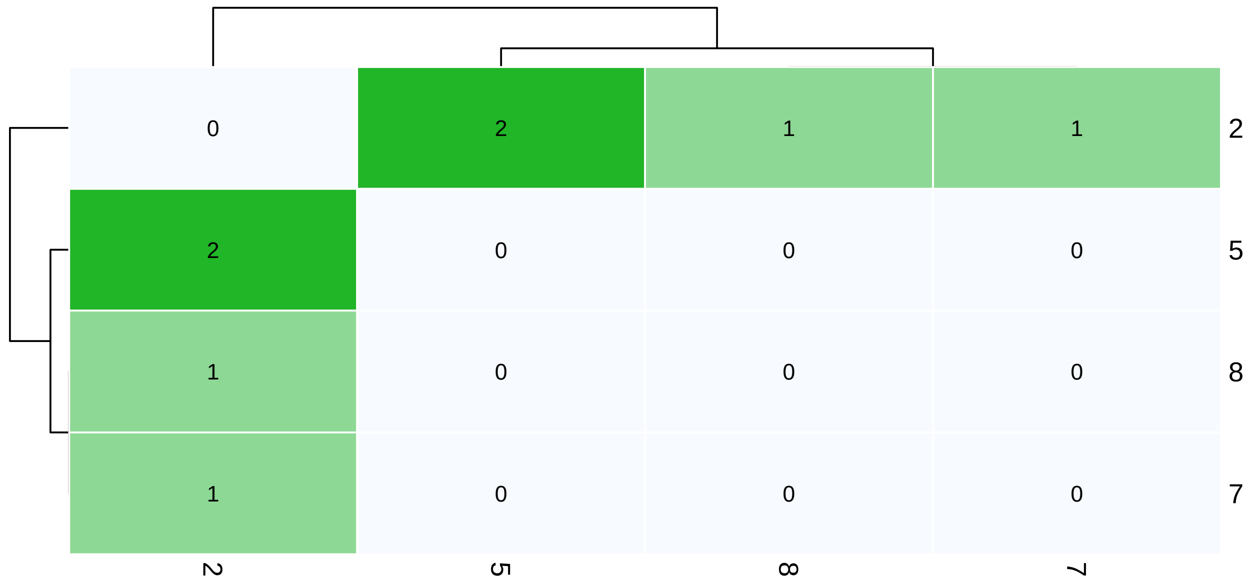

Figure 2: Pairwise SNP distance matrix for sequenced cases (L segment). Values are raw nucleotide differences.

Model Diagnostics

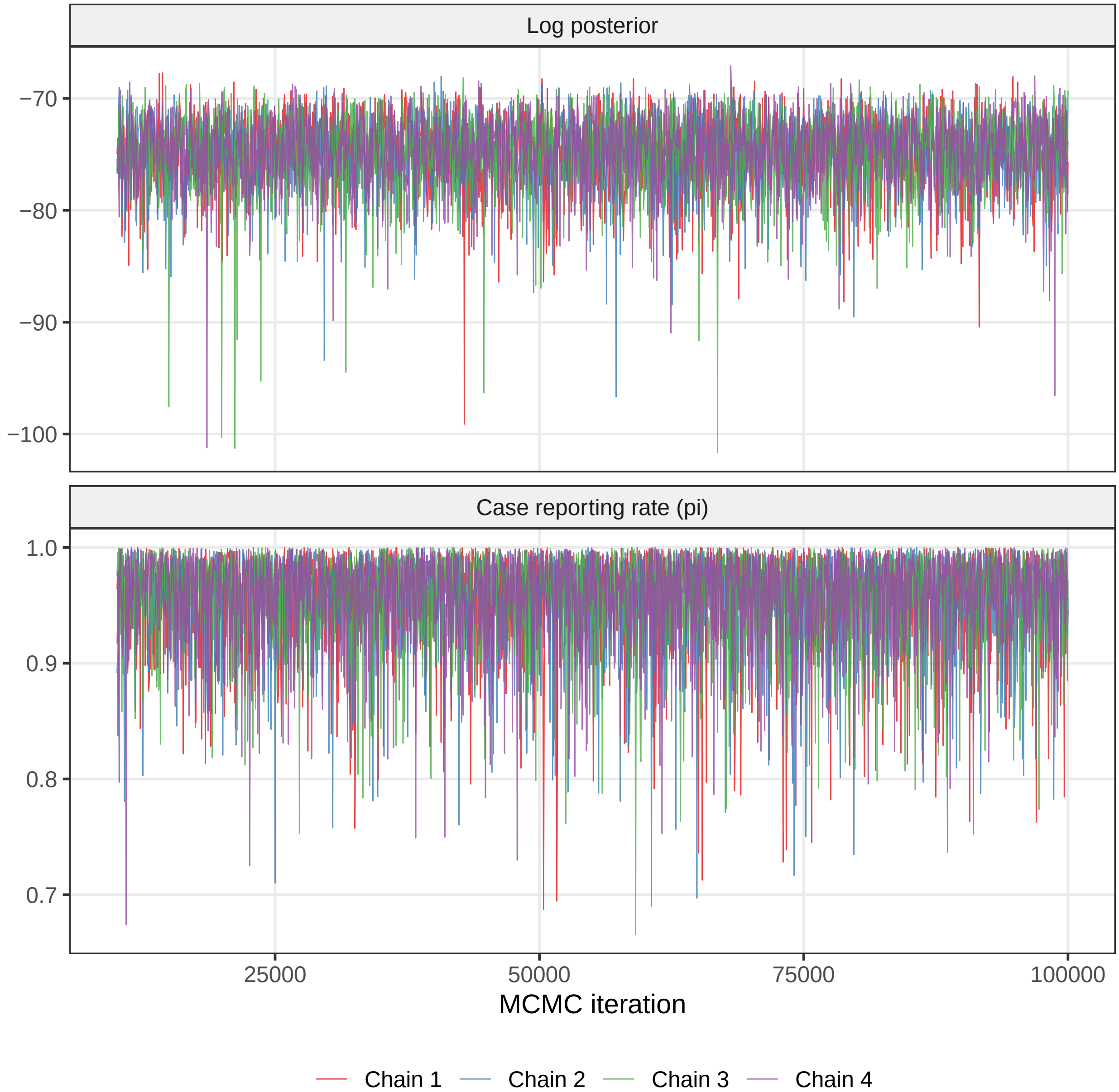

Figure 3: MCMC trace plots for scalar parameters (post-burnin). Stationary, well-mixed traces indicate convergence.

Table 1: Convergence diagnostics across 4 independent c

Parameter	R-hat
Case reporting rate (pi)	1.001
Infection times (t_inf, N = 13)	1.001

ESS: effective sample size out of 7200 total posterior samples (4 chains of 1800 samples each). The t_inf row sh

Transmission Tree

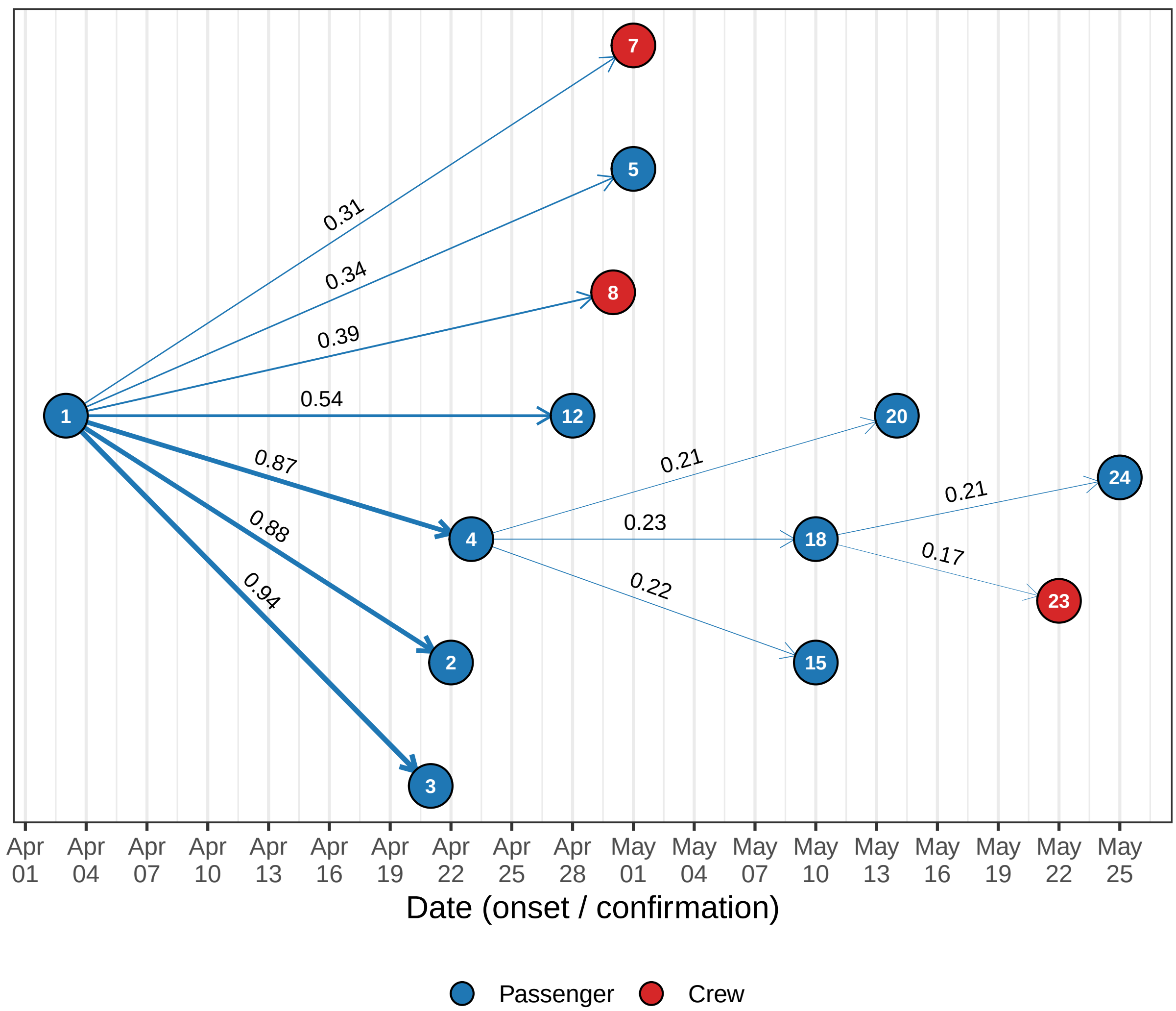

Figure 4: Consensus transmission tree (coloured edges), where each case is assigned its most frequently inferred ancestor across the posterior sample.

Ancestry Uncertainty

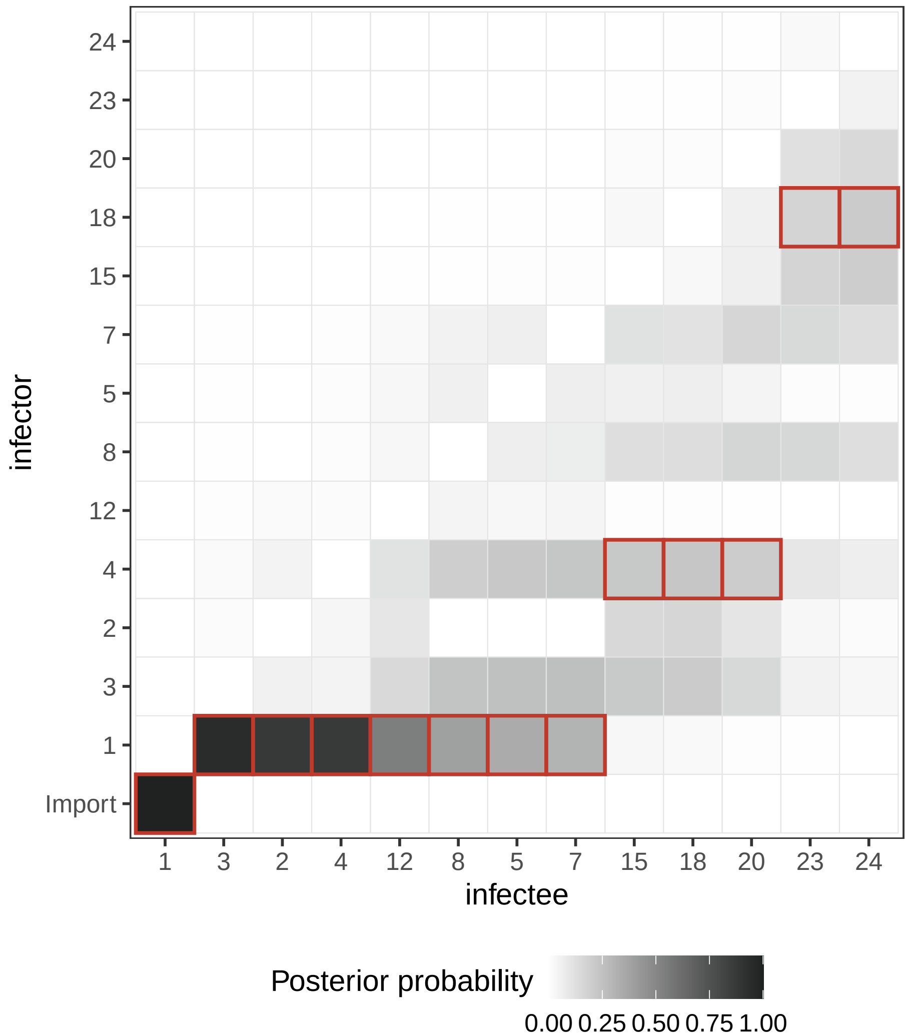

Figure 5: Ancestry support matrix: for each case (x-axis), the tile shading shows how often each potential infector (y-axis) was sampled across the posterior. Red outlines mark the consensus-tree ancestor (modal infector). 'Import' aggregates samples where the case was assigned no ancestor.

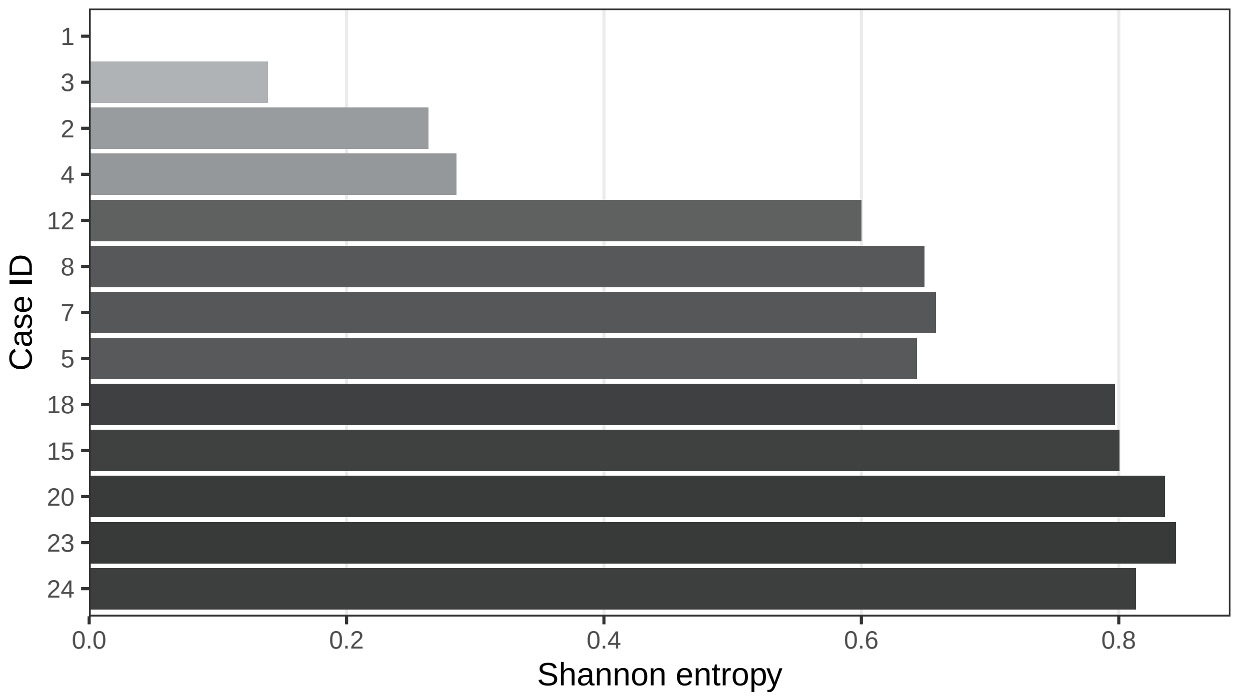

Figure 6: Shannon entropy of each case's inferred source. Entropy = 0 means a single unambiguous infector.

Dates of Infection

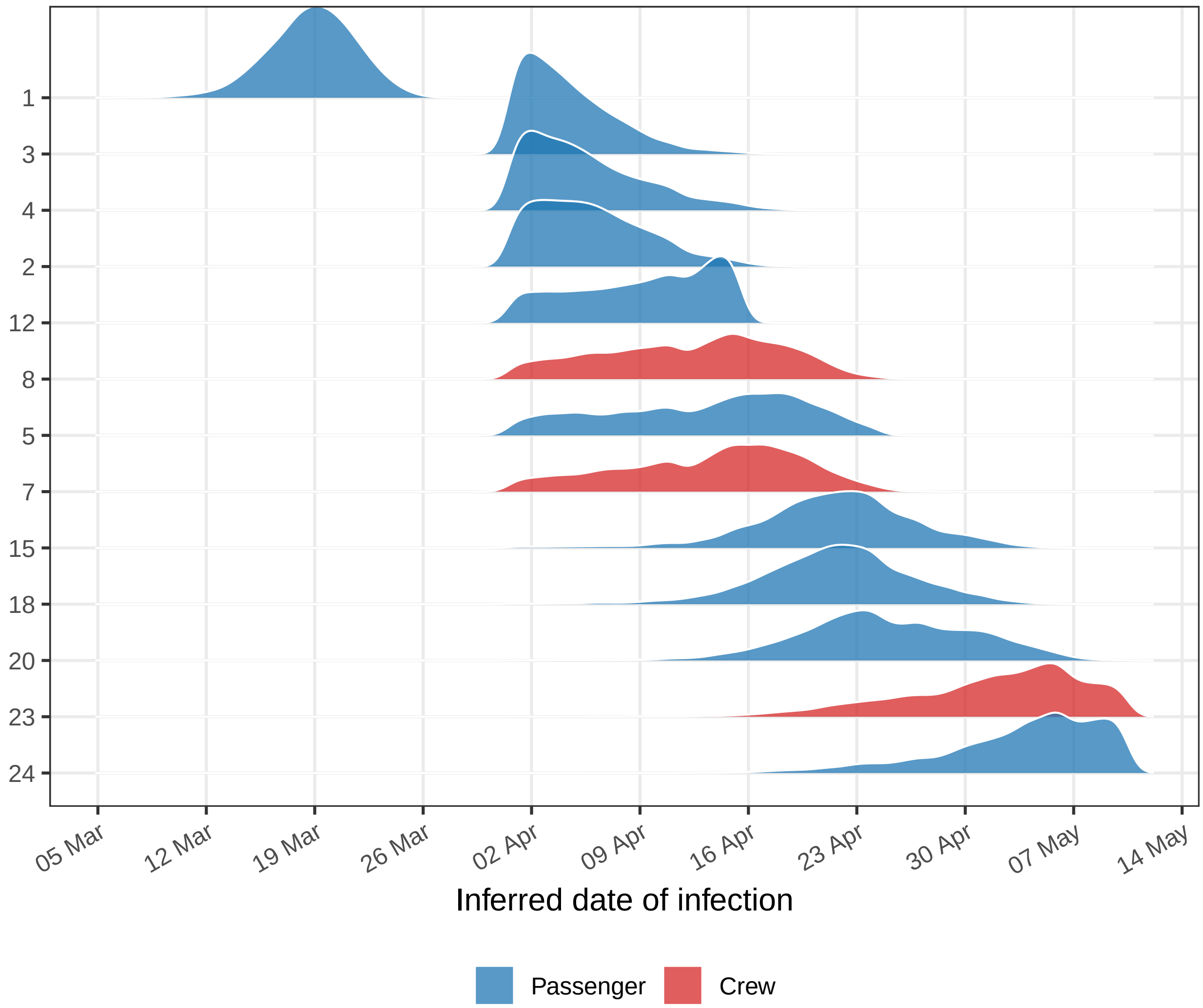

Figure 7: Posterior distribution of inferred infection dates per case. Width reflects uncertainty; cases ordered by mean inferred infection date.

Imports and Unobserved Transmission

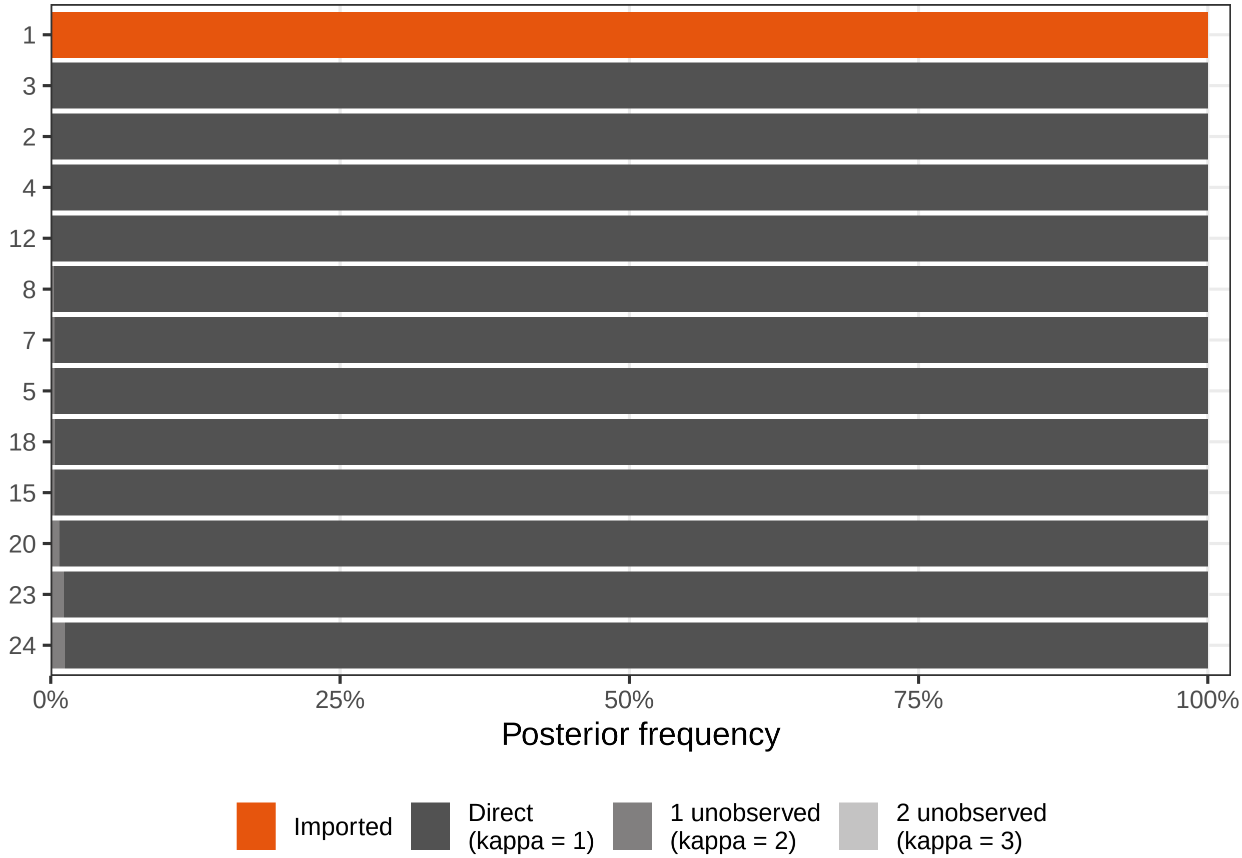

Figure 8: Posterior probability of each case being imported, directly linked, or connected via unobserved intermediate cases ( ).

Offspring Distribution

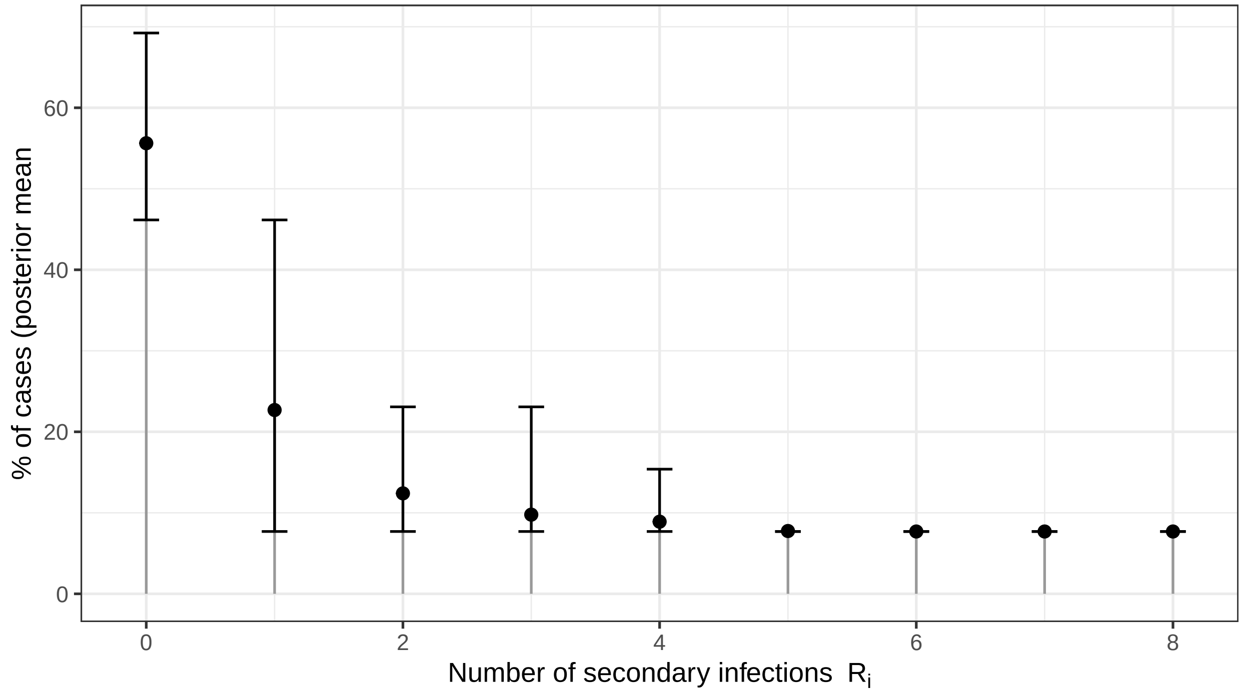

Figure 9: Posterior distribution of the offspring distribution, representing the distribution of secondary infections caused by an infected individual.

Discussion

The reconstruction is consistent with a single ANDV introduction onto the MV Hondius, with case 1 acting as a likely superspreader at the start of the voyage. Three of its onward links (cases 2, 3 and 4) are well supported (posterior 0.85); a further four (5, 7, 8, 12) are plausible but uncertain (0.31–0.54). The later wave, putatively seeded by case 4, sits at much lower support (0.17–0.23 for the edges to 15, 18, 20, 23 and 24) and should be read as suggestive rather than established.

Two factors limit resolution. First, cases 23 and 24 lack symptom-onset and ship-stay dates; using confirmation dates and a full-cohort exposure window almost certainly places them later in the tree than the true infection events. Second, the L-segment alignment carries near-zero genetic variation, so genomic data contribute little signal beyond confirming that the sequenced cases are co-clustered. The combined effect is high ancestry entropy (Fig. 6) for most cases after the initial cluster.

Despite this, the temporal and contact data are enough to support the broad picture: one introduction, one early superspreading event, and at most a small number of subsequent generations.

Updated linelist data, including onset dates and ship boarding and disembarkation dates for cases 23 and 24, would be valuable. Additional sequencing could help to further resolve the remaining ambiguous links.

Data

Data available under CC BY 4.0 at kraemer-lab/Hondius_hantavirus_h2026. Sequences retrieved from Pathoplexus. Report source: `report.qmd`.